

Roll No.

--	--	--	--	--	--	--	--

END SEMESTER EXAMINATION 2024

Name of the Course: B.Tech Petroleum Engineering

Semester: VII

Name of the Paper: Reservoir Management

Course Code: TPE-712

Time: Three Hours

MM: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions in each main question.
- iii. Total marks for each main question is **twenty**
- iv. Each Question carries **10 marks**

Q1		COs
a.	Write the various steps of reservoir Engineering and Reservoir Management. Explain MDT team and its function.	CO1
b.	What is the various technology required for Reservoir modeling & Simulation? Explain in detail.	CO1
c.	Explain mapping of a reservoir. What is the different process of reservoir mapping? Explain each in detail.	CO1
Q2		
a.	Explain the five important different phases of Oil & Gas field? Explain each in detail. Draw a cash flow diagram of an oil & Gas field with time.	CO2
b.	What are three different phases of development plan in view of recovery mechanism? What is the basic condition for employing secondary and tertiary recovery mechanism?	CO2 CO-3
c.	Design a Field Development Plan (FDP) with long- term production forecasts to support the business strategy and goals.	CO2 CO-4
Q3		
a.	How Geology play an important role in oil & Gas exploration? What are the essential elements of Petroleum Geology? Explain the term Source, Reservoir, Seal, Trap, Maturation, Pool with figures.	CO3 CO-4
b.	Explain Darcy law of permeability. Explain, Primary, secondary porosity, Facies, Permeability, effective and Relative permeability.	CO3 CO-4
c.	Explain Variogram with figure. How this is useful for petrophysical modeling process? Write the formula for variance calculation.	CO-3 CO-4
Q4		

a.	Explain Gas cap drive reservoir why gas cap drive reservoir is preferred? Draw a comparison diagram of Pressure Vs Time of both type of reservoir.	CO4
b.	What is the common cause for water production from an oil & gas field? What are the two different methods for water control? Explain with diagram.	CO4
c.	Explain EOR and when it is applied in the field? What is the essential requirement for EOR application? What are the different methods applied for achieving maximum oil from tertiary recovery?	CO4
Q5		
a.	Write the three different categories of Reserve. Explain each in detail in view of P90, P50 & P10. What are the deferent methods for reserve estimation?	CO4 CO5
b.	From logs, it has been determined that a productive formation has a bed thickness of 20 ft, a porosity of 25%, and a water saturation of 30%. The recovery factor is assumed to be 30%. The oil formation volume factor is 1.3. The areal extent of the reservoir is 200 acres. Find: - a. <i>HCPV</i> expressed in ft b. <i>OIP</i> expressed in bbl/acre c. Reserves expressed in stock-tank barrels	CO5
c.	What is the various factor responsible for Uncertainty in measurement? What are the SPE guidelines for E&P Companies for probability of Exceedance and non- Exceedance for oil and Gas assessment?	CO5